Student Assignment Brief

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Contents

Assignment Information

Module Name:	Introduction to Statistical Methods for Data Science
Module Code:	ST7089CEM
Assignment Title:	STW7089CEM Introduction to Statistical Methods for Data Science March Intake 2026 Coursework CW (Regular)
Assignment Due:	16-July-2026, 10:36PM

Assignment Credit:	15
Word Count:	6000
Assignment Type:	Coursework : Individual Report
Grading:	Percentage Grade (Applied Core Assessment).

Assessment Overview

You will be provided with an overall grade between 0% and 100%. You have one opportunity to pass the assignment at or above 40%.

Important Notice

The work you submit for this assignment must be your **own independent work**. More information is available in the 'Assignment Task' section of this assignment brief.

Assessed Module Learning Outcomes

The Learning Outcomes for this module align with the marking criteria which can be found at the end of this brief. Ensure you understand the marking criteria to ensure successful achievement of the assessment task. The following module learning outcomes are assessed in this task

1. Demonstrate knowledge of underlying concepts in probability and statistics used in Data Science.
2. Select and apply appropriate statistical methods or techniques to solve problems or analyse data sets.
3. Use modern software to solve real world problems and analyse large data sets.
4. Interpret the results of their analyses and communicate those results accurately.

Assignment Task

Coursework Title: Modeling EEG signals using polynomial regression

Task and Mark distribution:

Coursework Description:

The aim of this assignment is to select the best regression model (from a candidate set of nonlinear regression models) that can well describe the relationship between several ‘simulated’ brain electroencephalogram (EEG) signals. EEG is a widely used non-invasive

method to record electrical activity of the brain. Compared with other neuroimage data, such as functional magnetic resonance imaging (fMRI), EEG technique is much cheaper and has excellent temporal resolution. As a result, EEG-based analysis and modeling approaches have been extensively applied to characterize various neurological disorders (e.g., Parkinson’s disease, epilepsy, and tremor) and the development of brain-computer interfaces. To achieve those goals, it is very important to understand the connectivity between different brain areas, which can be obtained through the modeling and analysis of different EEG channels.
The ‘simulated’ 5 EEG time-series data are provided in the two separate Excel files. The first X.csv file contains 4 input EEG signals, X = {x1, x2, x3, x4}, respectively, and the second y.csv file contains the output EEG signal y. The file time.csv contains the sampling time of all EEG data in seconds.
All 5 EEG signals are subject to additive noise (assuming independent and identically distributed (“i.e.,”) Gaussian with zero-mean) with unknown variance.

Task 1: Preliminary data analysis

You should first perform an initial exploratory data analysis, by investigating:

1. Time series plots (of input and output EEG signals)
2. Distribution for each EEG signal
3. Correlation and scatter plots (between different input EEG signals and the output EEG) to examine their dependencies

Task 2: Regression: modeling the relationship between EEG signals.

We would like to determine a suitable mathematical model in explaining the relationship between the input EEG signals and the output signal, assuming such a relationship (brain connectivity) can be described by a polynomial regression model. Below are 5 candidate nonlinear polynomial regression models, and only one of them can ‘truly’ describe such a relationship. The objective is to identify this ‘true’ model from those candidate models following Tasks 2.1–2.6.
Candidate models are with the following structures:

$$\text{Model 1:} \quad y = \theta_1\, x_4 + \theta_2\, {x_1}^2 + \theta_3\, {x_1}^3 + \theta_4\, {x_2}^4 + \theta_5\, {x_1}^4 + \theta_{bias} + \varepsilon$$
$$\text{Model 2:} \quad y = \theta_1\, x_4 + \theta_2\, {x_1}^3 + \theta_3\, {x_3}^4 + \theta_{bias} + \varepsilon$$
$$\text{Model 3:} \quad y = \theta_1\, {x_3}^3 + \theta_2\, {x_3}^4 + \theta_{bias} + \varepsilon$$
$$\text{Model 4:} \quad y = \theta_1\, x_2 + \theta_2\, {x_1}^3 + \theta_4\, {x_3}^4 + \theta_{bias} + \varepsilon$$
$$\text{Model 5:} \quad y = \theta_1\, x_4 + \theta_2\, {x_1}^2 + \theta_3\, {x_1}^3 + \theta_4\, {x_3}^4 + \theta_{bias} + \varepsilon$$

Task 2.1:
Estimate model parameters $\boldsymbol{\theta} = \{\theta_1, \theta_2, \cdots, \theta_{bias}\}^T$ for every candidate model using Least Squares ($\widehat{\boldsymbol{\theta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$), using the provided input and output EEG datasets (use all the data for training).

Task 2.2:
Based on the estimated model parameters, compute the **model residual (error) sum of squared errors (RSS)**, for every candidate model.

$$RSS = \sum_{i=1}^{n} \left(y_i - \mathbf{x}_i\widehat{\boldsymbol{\theta}}\right)^2$$

Here $\mathbf{x}_i$ denotes the i^{th} row (i^{th} data sample) in the input data matrix $\mathbf{X}$, $\widehat{\boldsymbol{\theta}}$ is a column vector.

Task 2.3:
Compute the **log-likelihood function** for every candidate model:

$$\ln p\left(D|\widehat{\boldsymbol{\theta}}\right) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2}\mathrm{RSS}$$

Here, $\hat{\sigma}^2$ is the variance of a model's residuals (prediction errors) distributions $\hat{\sigma}^2 = \mathrm{RSS}/(n-1)$, with n the number of data samples.

Task 2.4:
Compute the **Akaike information criterion (AIC)** and **Bayesian information criterion (BIC)** for every candidate model:

$$AIC = 2k - 2\ln p\left(D|\widehat{\boldsymbol{\theta}}\right)$$
$$BIC = k \cdot \ln(n) - 2\ln p\left(D|\widehat{\boldsymbol{\theta}}\right)$$

Here $\ln p\left(D|\widehat{\boldsymbol{\theta}}\right)$ is the log-likelihood function obtained from Task 2.3 for each model, k is the number of estimated parameters in each candidate model.

Task 2.5:
Check the distribution of model prediction errors (residuals) for each candidate model. Plot the error distributions, and evaluate if those distributions are close to normal/Gaussian (as the output EEG has additive Gaussian noise), e.g., by using a Q-Q plot.

Task 2.6:
Select the 'best' regression model according to the AIC, BIC, and distribution of model residuals from the 5 candidate models, and explain why you would like to choose this specific model.

Task 2.7:
Split the input and output EEG dataset (X and y) into two parts: one part used to train the model, the other used for testing (e.g., 70% for training, 30% for testing). For the selected 'best' model, 1) estimate model parameters using the training dataset; 2) compute the model's output/prediction on the testing data; and 3) also compute the 95% (model prediction) confidence intervals and plot them (with error bars) together with the model prediction, as well as the testing data samples.

Task 3: Approximate Bayesian Computation (ABC)
Using the 'rejection ABC' method to compute the posterior distributions of the 'selected' regression model parameters in Task 2. 1) You only need to compute 2 parameter posterior distributions—the 2 parameters with the largest absolute value in your least squares estimation (Task 2.1) of the selected model. Fix all the other parameters in your model as constant by using the estimated values from Task 2.1. 2) Use a uniform distribution as a prior around the estimated parameter values for those 2 parameters (from Task 2.1). You will need to determine the range of the prior distribution. 3) Draw samples from the above uniform prior, and perform rejection ABC for those 2 parameters. 4) Plot the joint and marginal posterior distribution for those 2 parameters. 5) Explain your results.

Datasets:
time_1673241270748 (1).csv
X_1673241366257 (1).csvy_1673241374123 (1).csv

Submission Instructions

Requirement	Details
File Naming	NAME_studentID
File Format	.docx/.pdf format
Submission Method	Campus 4.0 platform (submission link provided 2 weeks before deadline)

Marking and Feedback

How will my assignment be marked?

Your assignment will be marked by the Module Team using standardized criteria.

How will I receive grades and feedback?

Provisional marks will be released once internally moderated. Feedback will be provided alongside grades release within 2 weeks (10 working days).

What will I be marked against?

Details of the marking criteria for this task can be found in the Assessment Marking Criteria section at the end of this brief.

Grade Requirements

You must achieve 40% or above to pass this assessment. Ensure you understand the marking criteria for successful completion.

Assignment Support and Academic Integrity

Getting Help

If you have any questions about this assignment, please meet your respective module leader or teacher for more information.

Language Standards

You are expected to use effective, accurate, and appropriate language within this assessment task.

Academic Integrity

The work you submit must be your own. All sources of information need to be acknowledged and attributed; therefore, you must provide references for all sources of information and acknowledge any tools used in the production of your work, **excluding Artificial Intelligence (AI)**. We use detection software and make routine checks for evidence of academic misconduct. Definitions of academic misconduct, including plagiarism, self-plagiarism, and collusion can be found in Student handbook in Campus 4.0. All cases of suspected academic misconduct are referred to for investigation, the outcomes of which can have profound consequences to your studies.

Support for Students with Disabilities

If you have a disability, long-term health condition, specific learning difference, mental health diagnosis or symptoms, contact the Student Support Office for assistance.

Unable to Submit on Time?

If events prevent you from submitting on time, guidance on extenuating circumstances is available in the Student Handbook or from the Student Support Office.

Administration of Assessment

Module Leader Name:	Shrawan Thakur
Module Leader Email:	stw0049@softwarica.edu.np
Assignment Category:	cw
Attempt Type:	regular

Component Code:	cw

Assessment Marking Criteria

Marking Scheme

This coursework is worth 15 credits (100%). This will be marked according to:

• 20% will be given for performing

Task 1: Preliminary data analysis (histogram plots, simple input-output correlation measures, time series plots, fitting linear model, ...). If you create any programming code, you must include this in the report.

• 40% will be given for performing Task 2: Regression: Task 2.1 and Task 2.6, 5% each; Task 2.7 has 10%. • 20% will be given to perform the rejection Approximate Bayesian computation (ABC) to compute the (approximated) posterior distribution of the regression model parameters.

• 10% will be given to appropriate discussion and interpretation of the results you obtained.

• 10% will be awarded for writing the report (around 3000-4000 words) in a structured, readable form and submitting the executable R scripts. The report should be in sections with appropriate headings and should include introduction, results, discussion, and conclusion sections.

• All your programming code should be included in the appendix of your report. Please display them in a structured way (put headings like Task 1, Task 2.3, etc.), with appropriate comments/annotations. You need to attach the original R code (or Python/Matlab), NOT the screenshots of the code. The code will be marked as part of the above marking scheme (for all the tasks in this coursework, you will need to provide the corresponding code; when you describe/discuss the tasks in the main text of the report, please reference the corresponding code section in the Appendix).

Mark allocation guidelines to students:

0-39	40-49	50-59	60-69	70+	80+
Work mainly incomplete and/or weaknesses in most areas	Most elements completed; weaknesses outweigh strengths.	Most elements are strong; minor weaknesses	Strengths in all elements	Most work exceeds the standard expected.	All work substantially exceeds the standard expected.

Marking Rubric Task 1 – Task 3 (80%)

<40%	40-49%	50-59%	60-69%	70+%
Little or no implementation of the tasks using R (or other programming languages) and the required approach. Did not	Some implementation of the tasks using R (or other programming languages), with or without use of the required	Good implementation of the tasks using the required approach and R (or other programming languages).	Very good implementation of the tasks using the required approach and R (or other programming languages).	Excellent implementation of the tasks using exactly the required approach and R (or other programming languages).

describe all the steps in a clear and structured way. Programming code is only partially or not included in the Appendix. It is not displayed in a structured way with explicit annotations. The code is not referenced appropriately in the main text. Some or few results are presented quantitatively. A lack of use of figures and tables.	approach. Partially described the steps of the implementation. Some programming code is included in the appendix. It is not displayed in a structured way or without explicit annotations. The code is not referenced appropriately in the main text. Some results are presented quantitatively, with or without the use of figures and tables.	Describe all the steps in a clear and structured way. All programming code is included in the Appendix, displayed in a structured way with clear annotations, and is referenced appropriately in the main text. Results are presented quantitatively and clearly with the use of figures and tables.	Describe all the steps in a clear and structured way. All programming code is included in the Appendix, displayed in a structured way with very clear annotations, and is referenced appropriately in the main text. Results are well presented quantitatively and clearly with the use of figures and tables.	Describe all the steps in a clear and structured way. All programming code is included in the Appendix, displayed in a structured way with excellent annotations, and is referenced accurately in the main text. Results are excellently presented and evaluated quantitatively, with the use of figures and tables.

Discussion and Interpretation (10%)

Little or no interpretation of the results without appropriate discussions and reflections.	Some interpretation of the results, but little in-depth discussion and reflection.	Good interpretation of the results, with appropriate discussions and reflections.	Very good interpretation of the results, with extensive discussions and reflections.	Excellent interpretation of the results, with in-depth discussions and reflections.

Report writing (10%)

The report is poorly written without a structured, readable format. A lack of clear presentation and interpretation of figures and tables.	The report is written in a readable format but without a clear structure. A lack of clear presentation and interpretation of figures and tables.	The report is written in a structured, readable format, with clear display and interpretation of figures/tables.	The report is well written in a structured, readable format, with clear display and interpretation of figures/tables.	The report has an excellent presentation. It is written in a structured, readable format, with apparent display and interpretation of figures/tables.